Group Project - Robot Time

Worth 25%

Due date: Tuesday 21st April

Purpose

You will work in groups of 2 (because I have limited hardware). At this stage you should know who your partner is and what day your lab time is scheduled.
You are required to create a system to intelligently control either the:
DJI Tello drone
https://www.youtube.com/watch?app=desktop&v=vDOkUHNdmKs&t=4m46s
or the
Lynxmotion Robotic Arm (4 DoF)
https://eu.robotshop.com/products/lynxmotion-lss-4-dof-robotic-arm-assembled?qd=43ca18d2d80e37031b62469f954cbbac

I have 6 Drones and 4 Arms and they have been assigned randomly. The main objective is to expose you to the realities of working with something real, controlling it intelligently, and all the challenges that presents. **You will only be able to work on the actual Robots during the assigned Lab times so make sure you attend all lab hours! Time passes very quickly.**

Setup

Both the DJI Tello and the Lynxmotion Robot Arm come with their own proprietary control systems which you will explore. However, you will be able to use Python libraries which provide access to the Robots' controls. You will be expected to make us of a camera as part of your project. The Drone has a built in one. The Arm does not, so you will use an external one (e.g. laptop camera) in that case.

To Do

You are to build a behaviour for your Robot that accomplishes some task, combining the use of the camera and the Robot movements.
A cool option would be a Robot behaviour that can be initiated/controlled visually by means of a camera. In other words, your system needs to be able to interpret body poses and/or hand gestures and get the Robot to perform an associated behaviour.
For gesture/pose recognition I would recommend the Mediapipe gesture recognition AI model: https://developers.googleblog.com/2021/09/drone-control-via-gestures-using-mediapipe-hands.html

You are free to invent your own behaviour(s). More complex behaviours will receive higher marks (see rubric below).

You must demonstrate your project in the last week of term.
You must also submit a short review of the project explaining:

1. **The process: what you did (including what you tried and did not work)**
2. **The Result: what was achieved**
3. **The Learnings: what did you take away from the project**

Demonstration

Groups will give a short demo of their projects in the last week of term on **Wednesday April 22nd** and **Friday April 24th** (Thursday is Industry showcase day).

Important Notes:

DJI Tello Drone - Needless to say but I will anyway, the drones can be dangerous. Although the rotors are small, they are sharp and deadly when turning at speed. Be absolutely vigilant at all times especially with other people nearby. Make sure you are connecting to the correct Drone over WiFi (there will be 6 in the room)! The drones can move fast! Always plan to have a **kill switch**. Manually turning the Tello upside down will automatically turn off all motors. Keep an eye on battery level. Low battery can result in odd behaviour. I will manage a battery bank during class.

Lynxmotion Robot Arm – These are expensive! **Lift them by the base board**. They are not industrial Robot Arms, so be gentle with them. That said, they are metal and can move quickly, so keep the area clear around them. There is a maximum range they can move in any direction. Trying to move them past this (which is easy to do by accident in code) results in a failsafe shutdown of the Arm, and you need to reboot it. There is a maximum weight they can lift (which is not very much).

Marking Scheme:

	Poor	**Ok**	**Good**	**Excellent**	**Weighting**
	0	**1-4**	**5-7**	**7-10**	
Demo					**70%**
Incorporates a Pose detection model or equivalent	Not working	It's there but not really used much	Works but only a couple of poses recognised. A single behaviour implemented.	Multiple poses recognised which are linked to multiple behaviours	15
Accuracy	Random	Does not really perform behaviours accurately	Performs behaviours with good accuracy	Evidence that behaviours are carried out with high accuracy in mind	15
Speed	Doesn't move	Slow as a wet week (relative to the Robot's ability)	Acceptable speed (relative to the Robot's ability)	High speed (relative to the Robot's ability)	15
Safety	None considered	Some but very rudimentary	Some clear evidence	Safety first design	15
Complexity	None	Very little e.g. Straight forward Robot movements	Some evidence of higher-level behaviours e.g. Robot moves to a particular object and/or picks up an object. Or Robot can locate a particular object and/or go to it.	High level of complexity. Robot performing behaviours that involve multiple input sensors and/or complex tasks e.g. Robot must locate a red ball, go to it, pick it up and move it somewhere. Or Robot must fly around searching for an object of a particular size and/or Colour etc.	40
Project Report	**Marks will depend on level of detail and quality of writing**				**30%**
-The Process					60
-The Result					20
-The Learnings					20

Note: Since this is a 2-person project, you will need to let me know what the division of labour was. But, if both people are happy with the effort of each other, just let me know that it was 50/50. If not then agree what the % split was and I will adjust the grading from the overall mark.